

STANDARD OPERATING PROCEDURE (SOP) FOR EQUIPMENT CLEANING IN A PHARMACEUTICAL MANUFACTURING UNIT

*A SELF-LEARNING PROJECT IN PHARMACEUTICAL QUALITY
ASSURANCE*



PREPARED BY
PARUL PAYASI
BACHELOR OF PHARMACY (B. PHARM)
DR. HARISINGH GOUR VISHWAVIDYALAYA, SAGAR (M.P.)

REFERENCE SOURCES
WHO GOOD MANUFACTURING PRACTICES (GMP) GUIDELINES
STANDARD PHARMACEUTICAL DOCUMENTATION PRACTICES
FACULTY OF PHARMACEUTICAL SCIENCES

SUBJECT AREA
PHARMACEUTICAL QUALITY ASSURANCE

ACADEMIC SESSION
2026–2027

VERSION 1.0
DATE: JULY 2026

DOCUMENT CODE:
SOP-QA-001

DOCUMENT CONTROL

Document Title	Standard Operating Procedure (SOP) for Equipment Cleaning in a Pharmaceutical Manufacturing Unit
Document Code	SOP-QA-001
Version	1.0
Effective Date	July 2026
Review Date	July 2027
Prepared By	Parul Payasi
Document Type	Self-Directed Academic Project
Department	Pharmaceutical Quality Assurance
Classification	Educational Document
Language	English
Total Pages	8

REVISION HISTORY

Version	Date	Description of Revision
1.0	July 2026	Initial version of the document

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1.0 Purpose

The purpose of this Standard Operating Procedure (SOP) is:

- To ensure proper cleaning of pharmaceutical equipment before and after manufacturing operations.
- To prevent cross-contamination by removing product residues, cleaning agent residues, dust, and microorganisms.
- To ensure compliance with Good Manufacturing Practices (GMP) and Schedule M requirements.
- To maintain the quality, safety, and efficacy of pharmaceutical products.
- To minimize the risk of contamination, product defects, and allergic reactions before manufacturing the next batch.

2.0 Scope

This SOP is applicable to:

- All pharmaceutical equipment used in manufacturing and production areas.
- Equipment cleaning activities performed before and after manufacturing operations.
- Production personnel responsible for equipment cleaning.
- Quality Assurance (QA) personnel responsible for verification and compliance.

3.0 Responsibilities

Production Operator

- Perform cleaning of equipment according to the approved SOP.
- Use approved cleaning agents and cleaning materials.
- Record and document all cleaning activities.

Production Supervisor

- Ensure that equipment cleaning is performed according to the SOP.
- Verify the cleanliness of the equipment.
- Review and sign the cleaning records before submission to QA.

Quality Assurance (QA)

- Verify that equipment cleaning complies with GMP and this SOP.
- Review the cleaning records for completeness and accuracy.
- Approve the equipment for further manufacturing after satisfactory verification.

4.0 Definitions and Abbreviations

4.1 Standard Operating Procedure (SOP)

A written document that provides standardized, step-by-step instructions for performing a specific activity consistently. It helps minimize errors and ensures

compliance with Good Manufacturing Practices (GMP).

4.2 Good Manufacturing Practices (GMP)

A system of guidelines and practices followed by the pharmaceutical industry to ensure pharmaceutical products are consistently manufactured with the required quality, safety, and efficacy.

4.3 Quality Assurance (QA)

A quality system that ensures pharmaceutical products are manufactured according to GMP through planning, documentation, monitoring, and review.

4.4 Cross-contamination

The unintended transfer of product residues, microorganisms, or other contaminants from one product, equipment, or surface to another, which may affect product quality and patient safety.

4.5 Product Residue

Any remaining pharmaceutical product or raw material left on the surface of equipment after manufacturing that must be removed before the next manufacturing batch to prevent cross-contamination.

4.6 Cleaning Agent

A validated detergent or cleaning material used to remove product residues, dust, dirt, and microorganisms from pharmaceutical equipment.

4.7 Abbreviations

Abbreviation	Full Form
SOP	Standard Operating Procedure
GMP	Good Manufacturing Practices
QA	Quality Assurance
QC	Quality Control
GDP	Good Documentation Practices
PPE	Personal Protective Equipment
WHO	World Health Organization
SS	Stainless Steel

5.0 Materials & Equipment

Cleaning Materials

- Approved detergent/cleaning agent
- Purified water
- Lint-free cloth
- Soft nylon brush

Personal Protective Equipment (PPE)

- Gloves
- Face mask
- Hair cover
- Safety goggles
- Apron/Lab coat

Documentation

- Equipment Cleaning Logbook
- Equipment Status Label

6.0 Procedure

6.1 Before Cleaning

- Wear the required PPE.
- Switch OFF and isolate the equipment from the power supply.
- Remove any remaining product residue.
- Ensure approved cleaning materials are available.
- Keep the Equipment Cleaning Logbook ready.
- Attach or verify the "To Be Cleaned" status label.

6.2 Cleaning Procedure

1. Remove all visible product residues.
2. Dismantle removable parts, where applicable.
3. Wash the equipment using the approved cleaning agent.
4. Clean difficult areas using a soft nylon brush.
5. Ensure all surfaces are thoroughly cleaned.
6. Prepare the equipment for rinsing.

6.3 Rinsing

1. Rinse the equipment thoroughly with purified water.
2. Ensure all detergent and product residues are removed.
3. Repeat rinsing if necessary.
4. Confirm no visible residue remains.

6.4 Drying

1. Allow excess water to drain.
2. Dry using a clean lint-free cloth or air drying.
3. Ensure no visible moisture remains.
4. Protect the equipment from contamination until use.

6.5 Visual Inspection

1. Visually inspect the equipment using the naked eye.

2. Ensure no visible product residue, detergent residue, dust, or stains remain.
3. Repeat cleaning if contamination is observed.
4. Submit the equipment for QA verification.

6.6 Documentation

1. Record cleaning activities in the Equipment Cleaning Logbook.
2. Document the equipment name/ID, date, time, and cleaning agent used.
3. Obtain the operator's and supervisor's signatures.
4. QA reviews and verifies the record.
5. Maintain records for audits and traceability.

6.7 Equipment Release

1. Submit the cleaned equipment and records to QA.
2. QA verifies equipment cleanliness and documentation.
3. Attach the "Cleaned" status label after approval.
4. Release the equipment for the next manufacturing process.

7.0 Precautions

- Ensure the equipment is switched OFF before cleaning.
- Wear the required PPE.
- Use only approved cleaning agents.
- Dismantle removable parts where applicable.
- Use clean lint-free cloths and approved brushes.
- Inspect the equipment for damage or loose parts.
- Ensure complete drying before use.
- Do not use the equipment without QA approval.
- Record all cleaning activities.

8.0 Documentation and Records

Maintain the following records:

- Equipment Cleaning Logbook
- Equipment Status Label Record
- Production Operator Signature Record
- Production Supervisor Review Record
- QA Verification and Approval Record

All records shall be maintained according to Good Documentation Practices (GDP).

9.0 References

- WHO Good Manufacturing Practices (GMP) Guidelines
- Schedule M, Drugs and Cosmetics Rules, 1945 (India)
- Good Documentation Practices (GDP)

- Standard Textbooks of Pharmaceutical Quality Assurance

10.0 Annexures

Definition

Annexures are supporting documents, forms, checklists, and templates attached to the SOP to record and demonstrate that the equipment cleaning procedure has been performed according to the approved SOP and Good Manufacturing Practices (GMP).

Annexure I – Equipment Cleaning Logbook

Date	Equipment Name/ID	Cleaning Used	Agent Cleaned By	Supervisor Signature	QA Approval	Remarks
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Annexure II – Equipment Status Label

TO BE CLEANED

Equipment Name: _____

- Equipment ID: _____
- Date: _____
- Operator Signature: _____

CLEANED

- Equipment Name: _____
- Equipment ID: _____
- Date: _____
- QA Signature: _____

Annexure III – Equipment Cleaning Checklist

- ☐ Equipment switched OFF
- ☐ PPE worn
- ☐ Product residue removed
- ☐ Approved cleaning agent used
- ☐ Equipment rinsed with purified water
- ☐ Equipment completely dried
- ☐ Visual inspection completed
- ☐ Documentation completed
- ☐ QA approval obtained